

Unit 2, Lesson 2: Square Values and Square Roots

Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^2 = q$, where p is a whole number and q is an integer, determine the real solutions.

Learning Targets

- ☐ I can determine the positive and negative solutions for $x^2 = p$, where p is a whole number.
- ☐ I can use the vocabulary related to radicals.
- ☐ I can interpret and evaluate a square root.

2.2.1 Warm-Up: Estimation

1. In 2006, scientists in North Wales found a deep-sea clam that was 507 years old. Deep-sea clams have the slowest growth rate, growing as little as 8.4 millimeters (mm), or one-third inch, every 100 years.



- a. Estimate the clam's size when the scientists found it in 2006.
- b. Explain your reasoning for your estimation.

2.2.2 Guided Instruction: Solving Equations in the Form $x^2 = p$

1. Consider the equation $x^2 = 1$.
 - a. Discuss with your partner what value(s) make the equation true.
 - b. Find the solutions for $x^2 = 1$.
 - c. Show the solutions that satisfy the equation.
2. Find the solutions for $64 = m^2$.
3. Find the solution for $x^2 = 0$.

For equations in the form $x^2 = p$, where p is a whole number, there are always two solutions, except in the case where $p = 0$. For all other whole number values of p , there is one positive and one negative solution.

4. Consider the equation $x^2 = -36$.

a. Discuss with your partner whether you think it is possible to find a solution to this equation.

b. Complete the statements.

There

☐ is
☐ is not

 a real number that can be multiplied by itself to result in -36 . Therefore,

- | |
|---|
| <ul style="list-style-type: none">☐ there is not a real solution to the equation $x^2 = -36$.☐ the real solution(s) to the equation $x^2 = -36$ is{are} _____. |
|---|

2.2.3 Try It: Solving Equations in the Form $x^2 = p$

Find the solution(s) for each equation.

1. $x^2 = 36$

2. $t^2 = 144$

3. $49 = n^2$

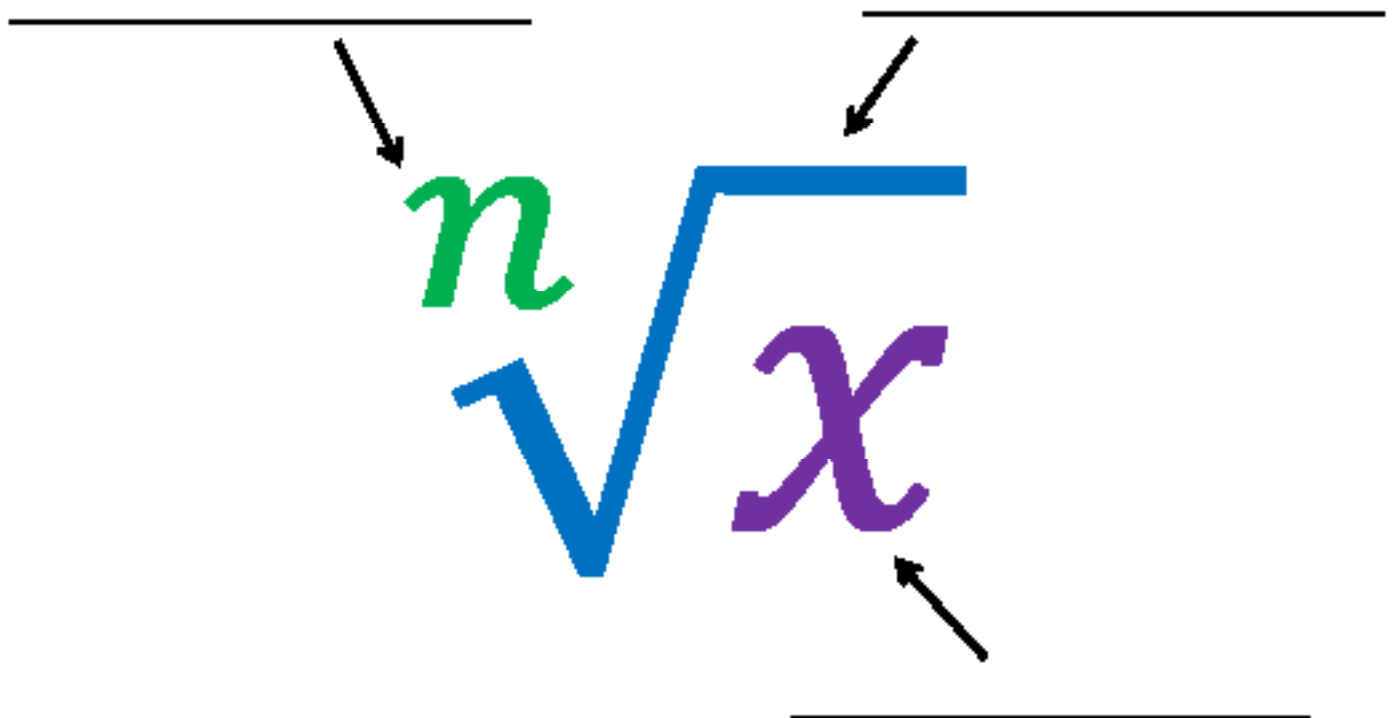
2.2.4 Guided Instruction: Square Roots

A perfect square is the product that results from squaring a number. Squaring a number is multiplying it by itself. The inverse operation to squaring a number is taking its **square root**.

The symbol $\sqrt{\quad}$ is called a **radical**. It is used to represent a **square root**. The value under the radical sign is called the **radicand**.

The **index** is the number outside the radical sign that indicates the root being taken.

1. Use this information to label the parts of a radical expression.



The symbol $\sqrt{\quad}$ is interpreted as the square root, unless an index other than 2 is indicated.

2. Consider the equations shown.

$$(4)^2 = 16$$

$$(-4)^2 = 16$$

The equations show which values can be squared to result in 16. Based on the information given about square roots, what is $\sqrt{16}$?

As explored in the previous lesson with the graph of $y = x^2$, every number has both a positive and negative square root, except for 0. The $\pm$ symbol is read, "plus or minus," and is used in front of the radical to denote both roots.

3. Complete each statement.

a. $\sqrt{4} = \pm 2$ because _____ x _____ = _____ and
_____ x _____ = _____

b. $\sqrt{121} = \pm$ _____ because _____ x _____ = _____
and _____ x _____ = _____

4. Explain why there is not a real number equivalent to $\sqrt{-81}$.

2.2.5 Guided Practice: Squares and Square Roots

1. Xavier completed two problems as shown.

Find the solution(s) for $b^2 = 25$.

$b = -5$ and $b = 5$

What is the value of $\sqrt{25}$?

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- a. Discuss Xavier's answers with your partner.

- b. Complete the statements.

Xavier answered ☐ one problem
☐ both problems correctly.

There ☐ is only one solution
☐ are two solutions to the equation

$b^2 = 25$ because there ☐ is only one value
☐ are two values that

make(s) the equation true. When finding the value of a square root,

consider what value can be multiplied by itself to result in the radicand.

Therefore, the value of $\sqrt{25}$ is ☐ 5.
☐ -5.
☐ ± 5 .

2. Find the value of each of the following.

a. $\sqrt{81} =$

b. $\sqrt{225} =$

c. $\sqrt{9} =$

d. $\sqrt{-64} =$

3. Solve each equation.

a. $a^2 = 169$

b. $225 = x^2$

2.2.6 Your Turn!

1. Find the value of each of the following.

a. $\sqrt{49} =$

b. $\sqrt{196} =$

2. Solve each equation.

a. $w^2 = 4$

b. $121 = k^2$

2.2.7 Wrap-Up: Reflection

- 1. Choose two statements to complete as you reflect on today's lesson.**
 - a. One thing that made the most sense to me today was ...**

 - b. One thing that I just don't understand is ...**

 - c. One thing I learned today was ...**

 - d. One thing I still wonder about is ...**

 - e. One thing I noticed about square roots is ...**

